good attempt

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National University of Computer and Emerging Sciences

FAST School of Computing   Fall-2022   Islamabad Campus

MT-1003: Calculus and Analytical Geometry (CS)

Friday, 11th November, 2022

Serial No:

Sessional Exam-II

Total Time: 1 Hour

Total Marks: 60

Signature of Invigilator

Course Instructors

Dr. Hamda Khan, Dr. Imran Shahzad, Dr. Naveed Khan, Mr. Ahtsham ul Haq, Mr. Muhammad Naseer

Aneeq Ahmed Malik — Student Name | 22i-1167 — Roll No. | BS-CS SB3 — Course Section | Student Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Attempt on question paper. Attempt all of them. Read the question carefully, understand the question, and then attempt it.
2. No additional sheet will be provided for rough work. Use the back of the last page for rough work.
3. If you need more space write on the back side of the paper and clearly mark question and part number etc.
4. After asked to commence the exam, please verify that you have nine (09) different printed pages including this title page. There are a total of 6 questions.
5. Calculator sharing is strictly prohibited.
6. Use permanent ink pens only. Any part done using soft pencil will not be marked and cannot be claimed for rechecking.
7. **Solve the questions using techniques learnt in this course.**

	Q-1	Q-2	Q-3	Q-4	Q-5	Q-6	Total
Marks Obtained	10	06	10	01	5.5	10	42.5
Total Marks	10	10	10	10	10	10	60

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Question 1 [10 Marks]

A farmer wants to construct animal feeder from a metal sheet. He wants to optimize the design of the feeder to avoid wasting any material. The design of the animal feeder is a hollow open-topped half cylinder. The radius of the semicircular ends is r cm and the length of the feeder is L cm. The metal used in the construction of the feeder is $600\pi\ cm^2$.

a) Sketch the shape of the animal feeder.

b) What is the constraint equation?

c) Write length L in terms of radius r.

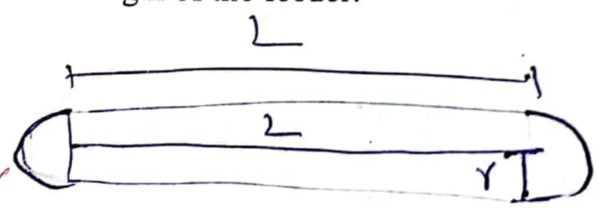
d) Write the equation of volume V in terms of r and L.

e) Write the equation of volume in terms of r alone.

f) Solve the optimization problem to determine the exact value of r for which V is optimized.

g) Verify that the value of r found in part (f) gives the maximum value for V.

h) Find, in exact form, the capacity and the length of the feeder.



Surface Area of feeder $= 600\pi\ cm^2$

$$S.A = \frac{\pi r^2}{2} + \frac{\pi r^2}{2} + \pi r L$$

$$S.A = \pi r^2 + \pi r L$$

As $S.A = 600\pi$

$$600\cancel{\pi} = \cancel{\pi}(r^2 + rL)$$

$600 = r(r+L) \Rightarrow \frac{600}{r^2} = 2r$

$600 = r^2 + 2r \Rightarrow \boxed{L = \frac{600 - r^2}{r}}$

V of half-cylinder $= \frac{1}{2}\pi r^2 L$

$V = \frac{1}{2}\pi r^2 \left(\frac{600 - r^2}{r}\right) \Rightarrow V = \frac{1}{2}\pi(600r - r^3)$

$V' = \frac{1}{2}\pi(600 - 3r^2)$ $\quad V'' = \frac{1}{2}\pi[0 - 6r]$

$V'' = -3\pi r$

$V' = 0 \Rightarrow 600 - 3r^2 = 0,$

$r = \pm 10\sqrt{2} \Rightarrow -10\sqrt{2}$ not part

$\boxed{r = 14.14}$

$V''\big|_{r=14.14} = -3\pi(14.14) < 0$, So

10

Value is maximum at $\boxed{r = 14.14\text{ cm}}$

$L = \frac{600 - r^2}{r} \Rightarrow \boxed{L = 28.29\text{cm}}$

$V = \frac{1}{2}\pi(600r - r^3) \Rightarrow V\big|_{r=14.14} = \boxed{8885.7\text{ cm}^3}$

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Question 2 [10 Marks]

In the engine shown below, a 7-inch connecting rod is fastened to a crank of radius 3 inches. The crankshaft rotates counterclockwise at a constant rate of 200 revolutions per minute.

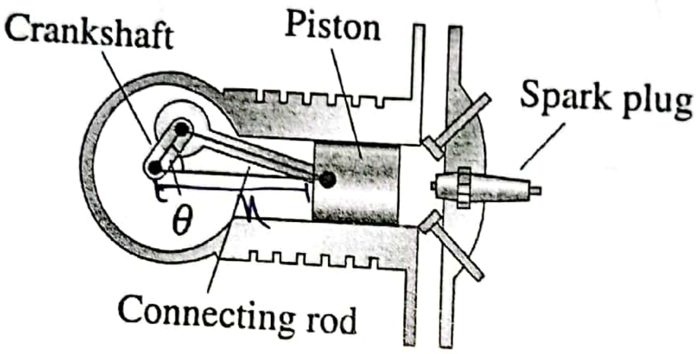
a) Write the expression for rate of change of θ.

b) Find an equation that relates x and θ.

c) Find a differential equation for rate of change of position of the piston x.

d) Find position x for $\theta = \frac{\pi}{3}$.

e) Find the velocity of the piston at $\theta = \frac{\pi}{3}$.



$$\frac{d\theta}{dt} = 200(2\pi r)/min$$

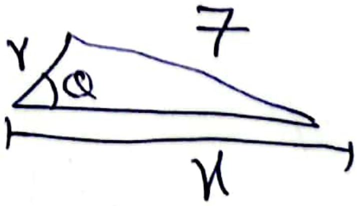
$$\frac{d\theta}{dt} = 400\pi r/min$$

As $r = 3$

06

$$\frac{d\theta}{dt} = \cancel{400}\ 1200\pi/min$$

$$\boxed{\tan\theta = \frac{7}{x}} \quad \text{—} (1)$$



differ by eq — (1)

$$\sec^2\theta \frac{d\theta}{dt} = \frac{-7}{x^2} \cdot \frac{dx}{dt} \quad (A)$$

from (1), when $\theta = \frac{\pi}{3}$

$$\tan\frac{\pi}{3} = \frac{7}{x}$$

$$\sqrt{3} = \frac{7}{x} \Rightarrow \boxed{x = 4.04 \text{ inch}}$$

$$\left.\frac{dx}{dt}\right|_{\substack{\theta=\pi/3 \\ x=4.04}} = ?$$

$$\sec^2\left(\frac{\pi}{3}\right)(1200\pi) = \frac{-7}{(4.04)^2} \cdot \frac{dx}{dt}$$

$$4(1200\pi) = \frac{-7}{16.32} \frac{dx}{dt}$$

$$\frac{78336\pi}{-7} = \frac{dx}{dt}$$

Consult Sol

$$\boxed{\frac{dx}{dt} = -35157 \text{ inch/min}} \Rightarrow \cancel{\frac{dx}{dt} = 585}$$

$$\boxed{\frac{dx}{dt} = -585.9 \text{ inch/sec}}$$

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Question 3 [10 Marks]

Find a function f such that $f'(x) = x^3$ and the line $x + y = 0$ is tangent to the graph of f.

line $\Rightarrow x+y=0 \Rightarrow y=-x$

Slope of tangent line $= -1$ (A)

As $f'(x) = x^3$ —— (1)

Equation (1) is the general slope formula for all tangents that can be drawn at f. $f'(x) = -1$ from (A)

$x^3 = -1$

$\boxed{x = -1}$ other roots are imaginary

When $\boxed{x=-1}$, the y will be

$y = -(-1) \Rightarrow \boxed{y = 1}$

The tangent $y+x=0$, cuts the graph of f, at $(-1,1)$

from this part, we have the
Initial Condition

$$f(-1) = 1 \quad \text{— } B$$

$$f'(x) = x^3$$

Integrating both sides

$$\int f'(x) = \int x^3$$

$$f(x) = \frac{x^4}{4} + c$$

As

$$f(-1) = 1$$

$$1 = \frac{1}{4} + c \Rightarrow c = 1 - \frac{1}{4}$$

$$\boxed{c = \frac{3}{4}}$$

10

So,

$$\boxed{f(x) = \frac{x^4}{4} + \frac{3}{4}}$$

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Question 4 [10 Marks]

Find the derivative $\frac{dy}{dx}$ of the implicit function $x^y + y^x = c$.

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$\Rightarrow x^y + y^x = c$

Taking natural log on both sides.

$\ln(x^y + y^x) = \ln c$

$\ln x^y + \log y^x = \ln c$

$y \ln x + x \ln y = \ln c$

differentiating both sides w.r.t x

$y \cdot \frac{1}{x} + \ln x \frac{dy}{dx} + \frac{x}{y}\frac{dy}{dx} + \ln y = 0$

$\frac{dy}{dx}\left[\ln x + \frac{x}{y}\right] + \ln y + \frac{y}{x} = 0$

$\frac{dy}{dx}\left[\frac{y \ln x + x}{y}\right] + \frac{xy + \ln y}{x} = 0$

$\frac{dy}{dx} = -\frac{xy + \ln y}{x} \times \frac{y}{y \ln x + x}$

$\boxed{\frac{dy}{dx} = -\frac{xy + \ln y}{y \ln x + x} \cdot \frac{y}{x}}$

Question 5A [4 Marks]

A horse is carrying a carriage on dirt path. The amount of energy E (in calories) expended by the horse depends on the distance m (in miles) the horse walks. Also, the distance walked by the horse depends on time t (in hours). If the horse expends 40 calories per mile and the horse walks at a speed of 8 mph, at what rate $\frac{dE}{dt}$ is the horse expending energy.

$E \propto m$ (Given)

$\frac{dE}{dm} = 40$ calor/mile $\quad \frac{dm}{dt} = 8$ mil/h

$\cancel{E = mt}$

differenciating both sides w.r.t t

$\cancel{\frac{dE}{dt} = \frac{dm}{dt}}$ By Chain Rule

$$\frac{dE}{dt} = \frac{dE}{dm} \times \frac{dm}{dt}$$

$$\frac{dE}{dt} = 40 \times 8$$

04

$$\boxed{\frac{dE}{dt} = 320 \text{ calories/hour}}$$

Question 5B [6 Marks]

Suppose that t months from now the population of a town will be growing at the rate of

$$g(t) = 50 + 10\sqrt{t}$$

people per month. The initial population is 5000. Use the net change theorem to find the population in 3 years.

$g(t) = 50 + 10\sqrt{t}$

Integrating both sides, now, ftn be p(u)

$p(t) \Rightarrow 50t + 10 \cdot \frac{2}{3} t^{3/2} + c$

$p(t) \Rightarrow 50t + \frac{20}{3} t^{3/2} + C$

Initial Condition $\Rightarrow$ 5000 at $t = 0$

1.5

$p(0) = 5000$

$5000 = c \Rightarrow$

$p(t) = 50t + \frac{20}{3} t^{3/2} + 5000$

Now,

$p(3) = 50(3) + \frac{20}{3} (3)^{3/2} + 5000$

$p(3) = 5184$ (On Next page) with Net theore

Net change theorem

$$g(t) = \int_0^3 50 + 10\sqrt{t}$$

$$= \left[50t + \frac{20}{3}t^{3/2}\right]_0^3$$

$$= \left[50(3) + \frac{20}{3}(3)^{3/2}\right]$$

$$\Rightarrow 184.6$$

After three years, population will have an increase of 184,

Initial Population = 5000

After three years = 5184

Question 6A [4 Marks]

Find the derivative of following function by using Fundamental Theorem of Calculus

$$y = \int_{-1}^{x} \frac{t^2}{t^2+4}\,dt + \int_{x}^{3} \frac{t^2}{t^2+4}\,dt.$$

Using direct method,

$$y = \int_{-1}^{x} \frac{t^2}{t^2+4}\,dt + \int_{x}^{3} \frac{t^2}{t^2+4}\,dt$$

$$y = \int_{-1}^{x} \frac{t^2}{t^2+4} - \int_{3}^{x} \frac{t^2}{t^2+4}\,dt$$

Taking derivation on both sides

$$\frac{dy}{dx} = \frac{d}{dx}\int_{-1}^{x} \frac{t^2}{t^2+4}\,dt - \int_{3}^{x} \frac{t^2}{t^2+4}\,dt$$

$$\frac{dy}{dx} = \frac{x^2}{x^2+4} - \frac{x^2}{x^2+4}$$

$$\boxed{\frac{dy}{dx} = 0}$$

0

2

Question 6B [6 Marks]

If $f(x) = \frac{1}{3}x^3 - x^2 - 3x + 4$, Discuss the local extrema, point of inflection, and sketch the graph.

$f(x) = \frac{1}{3}x^3 - x^2 - 3x + 4$ x-intercept $(-2.5,0)$ $(4.4,0)$ $(1,0)$

Dom: $(-\infty,\infty)$ y-intercept: $(0,4)$

Symmetry: No Symmetry Asymptote: No Asymptote

$f'(x) = \frac{1}{3} \cdot 3x^2 - 2x - 3$

$\boxed{f'(x) = x^2 - 2x - 3}$

$\boxed{f''(x) = 2x - 2}$

Inc/Dec Interval:

Critical Pt's $f'(x) = 0 \Rightarrow x^2 - 2x - 3 = 0$

$\boxed{x = 3}$ & $\boxed{x = -1}$ $\Rightarrow (3, -5)$ $\Rightarrow (-1, 5.67)$

Interval	$(-\infty,-1)$	$(-1,3)$	$(3,\infty)$
Sign of f'	+ve	-ve	+ve
Behavior of f	Inc	dec	Inc

Maxima $\Rightarrow (-1, 5.67)$
Minima $\Rightarrow (3, -5)$

Concavity

Pt's of Inflexion $f''(x)=0$

$2x=2 \Rightarrow x=1$

Interval	$(-\infty,1)$	$(1,\infty)$
Behavior of f	$\cap$ Concave down	$\cup$ Conc up
Sign of f''	$-$ve	+ve

pt. of Inflexion $\Rightarrow$ $(1, 0.33)$